INFOSYS SPRINGBOARD AI INTERNSHIP

Multi-Agent AI Analysis System for Financial Research and Business Insights

An evidence-grounded agentic platform for corporate annual report analysis and competitor benchmarking

INTERN / DEVELOPER

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Infosys Springboard Virtual Internship

PROJECT DOMAIN

Agentic AI & Financial Analytics

Document Extraction, & Risk Detection

MENTORSHIP & DATES

[Infosys Mentor]

Internship Year: 2026

Problem Statement & Project Objectives

THE PROBLEM IN CORPORATE RESEARCH

- **Information Overload:** Corporate annual reports and Form 10-K filings often exceed 300-500 pages, making manual analysis slow and error-prone.
- **Hallucination Risks in Generic LLMs:** Traditional AI chatbots invent figures, misread OCR digits, and cannot be trusted for financial decision-making.
- **Reporting Basis Confusion:** Standard tools frequently mix consolidated (group-level) and standalone (parent-only) figures without audit trails.
- **Absence of Traceability:** Analysts require verifiable citations (page numbers, chunk text) before accepting any extracted metric.

PRIMARY PROJECT OBJECTIVES

- **Automate Financial Ingestion:** Ingest corporate PDF filings, extract high-fidelity text with PyMuPDF, and index segments in ChromaDB.
- **Deterministic Metric Extraction:** Reliably capture Revenue from Operations and Net Profit across reported financial years.
- **Strict Document Isolation & Grounding:** Answer questions using verifiable RAG where 100% of claims cite exact PDF page numbers and chunk IDs.
- **Automated Financial Red-Flag Detection:** Flag year-over-year revenue contractions with deterministic severity levels (<20% vs >=20%).
- **Cross-Company Benchmarking:** Align metrics across companies (e.g. Reliance vs. Infosys vs. TCS) and calculate absolute and percentage variances.

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System Architecture & End-to-End Workflow

1. INGESTION

PDF Upload
PyMuPDF (fitz)
Text Extraction

2. CHUNKING

Overlapping Segments
500-char Chunks
Page Metadata

3. EMBEDDING

all-MiniLM-L6-v2
384-dim Vectors
ChromaDB Store

4. EXTRACTION

P&L Statement Rules
Revenue & Profit
Consolidated/Standalone

5. AGENT LAYER

Research Agent
Red Flag Agent
Comparison Agent

6. SYNTHESIS

Executive Summary
Key Financials
Streamlit UI & API

DATA STORAGE & RETRIEVAL LAYER

- SQLite (SQLAlchemy): Stores session state and document records.
- ChromaDB Vector Store: Stores document chunks with metadata filtering (`document_id`, `page_number`).
- Deterministic Query Routing: Filters by document ID to strictly prevent cross-company data leakage.

DETERMINISTIC PROCESSING LAYER

- Python Arithmetic: All percentages and variances are calculated mathematically, avoiding generative LLM arithmetic errors.
- Verified P&L Anchoring: Net profit extraction reuses verified P&L pages identified during revenue statement parsing.
- Traceability: Every finding preserves PDF page numbers and source excerpts.

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Multi-Agent System: Specialized Roles & Interactions

DOCUMENT AGENT
Parsing & Vector Indexing

Extracts raw text page-by-page from PDFs using PyMuPDF, chunks content into overlapping segments, generates embeddings with all-MiniLM-L6-v2, and indexes vectors into ChromaDB.

Module: services/parser.py, chunker.py, embedder.py

EXTRACTION AGENT
Financial Table Parsing

Identifies formal P&L statement pages, extracts Revenue from Operations across reporting periods, captures currency units (crores/lakhs), and handles OCR character replacements.

Module: agents/extraction_agent.py

RED FLAG AGENT
Automated Risk Detection

Evaluates year-over-year revenue trends. Detects revenue contractions and assigns deterministic severity ratings (Medium: <20% decline, High: >=20% decline) with source citations.

Module: agents/red_flag_agent.py

RESEARCH AGENT
Cited Financial Q&A

Processes natural language financial inquiries, executes isolated ChromaDB vector retrieval, synthesizes verified answers, and attaches exact page numbers and chunk excerpts.

Module: agents/research_agent.py

COMPARISON AGENT
Cross-Company Benchmarking

Aligns equivalent metrics across two or more corporate reports, enforces unit/reporting basis compatibility, and computes absolute, signed, and percentage differences.

Module: agents/comparison_agent.py

REPORT AGENT
Executive Report Synthesis

Orchestrates outputs from Extraction, Red Flag, and Comparison agents to compile an initial financial report featuring an Executive Summary and Key Financials table.

Module: agents/report_agent.py

Technology Stack & Technical Decisions

BACKEND ARCHITECTURE

- **Python 3.12:** Chosen for modern type annotations, high-speed execution, and AI library compatibility.
- **FastAPI & Uvicorn:** High-performance asynchronous REST API framework with native Pydantic validation and auto-generated Swagger UI.
- **SQLite & SQLAlchemy:** Zero-configuration relational database managing research sessions and uploaded document metadata.

VECTOR RETRIEVAL & PARSING

- **PyMuPDF (fitz):** High-speed PDF text parsing that preserves layout coordinates and exact page boundaries.
- **ChromaDB:** Open-source, local vector database supporting metadata filtering (`document_id`, `page_number`).
- **SentenceTransformers (all-MiniLM-L6-v2):** Fast, lightweight (80MB) local embedding model generating 384-dimensional dense vectors on CPU with zero API costs.

FRONTEND & TESTING SUITE

- **Streamlit:** Modern data application interface providing interactive chat, comparative tables, and file uploaders.
- **Requests Client:** Decoupled communication layer (`api_client.py`) connecting Streamlit to the frozen FastAPI backend.
- **Python unittest:** Rigorous test suite comprising 60 automated unit and integration tests (100% passing).

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Implementation Results: Validated Financial Outputs

CASE STUDY DATASET: Reliance Industries Limited (Integrated Annual Report 2024-25)
Validated across Conversational Research, Red Flag Detection, and Executive Reporting

1. CONSOLIDATED REVENUE

Rs. 980,136 Cr

FY 2024-25 vs Rs. 914,472 Cr in FY 2023-24
Growth: +7.18% (Calculated via Python)

Verified Citation:
- Document: Reliance Annual Report 24-25
- Page: 100 (Consolidated Statement)
- Chunk: ..._p100_c866

2. RED FLAG DETECTION

RF001: Medium Risk

Standalone Revenue: Rs. 532,792 Cr (FY25)
vs Rs. 547,942 Cr (FY24)
Decline: -2.76% (Triggered warning)

Verified Citation:
- Document: Reliance Annual Report 24-25
- Page: 58 (Standalone Statement)
- Chunk: ..._p58_c546

3. EXECUTIVE REPORT AGENT

Initial Report

Synthesizes Key Financials & Narrative:
- Revenue: Rs. 980,136 Cr (+7.18%)
- Net Profit: Rs. 80,787 Cr (+2.74%)
- Citations preserved intact

Competitive Benchmarking:
- Reliance vs. Infosys (Rs. 162,990 Cr)
  Abs Diff: Rs. 817,146 Cr (-83.37%)
- TCS Comparison: Rs. 255,324 Cr (P. 181)

Project Limitations & Future Enhancements

CURRENT SYSTEM LIMITATIONS

- **Scoped Financial Metrics:** Current extraction focuses on core Income Statement figures (Revenue and Net Profit); Balance Sheet items (Total Debt, Cash Flow) require dedicated extractor patterns.
- **Text-Based PDF Dependency:** Relies on text layers extracted by PyMuPDF; scanned image-only PDFs without an embedded OCR layer require external pre-processing.
- **Initial Report Structure:** Report Agent produces Executive Summaries and Key Financials; long-form segment breakdown and audit note chapters are deferred.
- **Local Embedding Latency:** Indexing massive 400+ page annual reports requires ~25-40 seconds on local multi-core CPU.

FUTURE ENHANCEMENTS & ROADMAP

- **Deep Learning Table Extraction:** Integrate vision-based table transformers (e.g. Table Transformer / LayoutLM) to parse complex bordered/unbordered tables.
- **Expanded Metric Coverage:** Broaden automated extraction to include Operating Margins, EBITDA, Earnings Per Share (EPS), and Free Cash Flow.
- **Automated PDF Export:** Add a one-click downloadable PDF report generator featuring automated financial charts and trend visualizations.
- **Enterprise Scalability:** Transition from SQLite to PostgreSQL and deploy ChromaDB as a distributed vector cluster for multi-user concurrency.

Conclusion & Project Summary

KEY TAKEAWAYS & PROJECT IMPACT

✓ **Deterministic Accuracy Over Generative Guesswork:** By coupling vector retrieval with exact Python arithmetic, the system delivers zero-hallucination calculations for revenue, profit, and percentage changes.

✓ **Full Source Traceability:** Every factual statement links directly to its source PDF page number, chunk ID, and statement type, establishing complete audit confidence.

✓ **Specialized Multi-Agent Synergy:** Dividing responsibilities across Document, Extraction, Red Flag, Research, and Comparison agents ensures modularity, maintainability, and clean testing.

✓ **Proven Software Engineering Standards:** Validated with 60 comprehensive unit and integration tests, clean REST API endpoints in FastAPI, and an intuitive Streamlit research interface.

Thank you! Open for Questions & Viva Discussion.